Avani Chouhan

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.21
Ideal Public School, Balaghat M.P.	Class XII (CBSE)	2023	70.00%
Ideal Public School, Balaghat M.P.	Class X (CBSE)	2021	82.00%

SKILLS

Language Proficiency: English, Hindi, Marathi

EDA & Circuit Design: Cadence Virtuoso, LTspice, KiCad, Altium Designer

HDL & Programming: Verilog, C, MATLAB, Bash/Shell Scripting, SQL

Documentation & Design: LaTeX, Solid Edge

RELEVANT COURSES

Theory Courses: Digital Systems, Circuits & Network Analysis, Control Theory for Circuits, CMOS Processing, Microprocessors & Computer Architecture, Digital IC Design; Introduction to HDL, RTL Design & Verification, Memory Circuits & System Design, Memory Device Technology for AI/ML, Productivity Tools for IC Design, Current Trends in EE.

Lab Courses: Electrical Circuits Lab, Digital Systems Lab, Semiconductor Devices Lab, Electronic System Design Project Lab, IC Design Test & Characterization.

Programming Courses: Introduction to Programming

PROJECTS

Digital Stethoscope

- Developed a digital stethoscope using a condenser microphone and ATmega328P for real-time heart and lung sound capture and processing.
- Designed a custom PCB in KiCad and implemented an amplifier circuit to improve signal quality before ADC conversion.
- Integrated a TFT LCD to display real-time signal waveforms.
- Designed and 3D-printed a compact enclosure using Solid Edge and Ultimaker Cura.

RTL Synthesis and Benchmark Analysis

- Synthesized and analyzed RTL designs and benchmarks (ISCAS-85, CEP, ITC99, and UART) using Cadence Genus as part of the Productivity Tools for IC Design course.
- Compared area, timing, and power metrics across different process corners and operating frequencies.
- Automated design execution and report parsing using TCL scripting, improving synthesis workflow efficiency and reducing manual analysis.

CORDIC-Based K-Means Implementation

- Designed a low-complexity K-Means clustering architecture using a CORDIC engine to reduce computational overhead in Verilog.
- Generalized 2D CORDIC-based Euclidean distance computation to n-dimensional vectors, enabling a recursive and hardware-reusable architecture.

POSITIONS OF RESPONSIBILITY

UG Sunshine Buddy, IIT Hyderabad: Served as a peer-support buddy for undergraduate students, fostering open

communication and helping create a safe and supportive environment for students to discuss personal and academic concerns.

ELAN & nVision, IIT Hyderabad: Volunteered in Hospitality and Cult for the institute's annual cultural and technical fest; served as Deputy Coordinator for Beggies, assisting in planning and execution of fest activities.

Managed coordination between volunteers and participants, contributing to the smooth execution of events and handling on-ground responsibilities during the fest.

EXTRACURRICULAR

Inter IIT Cultural Meet : Represented IIT Hyderabad in Group Dance as a Core member of the dance club, secured top 10th position.

Interests : Reading, Running(NSO iith), Dancing and Baking.